Demo: Kelvin Water Dropper

ChatGPT edition — adult-built apparatus; measured evidence; stop early

Prove electrostatic feedback without making a spark

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

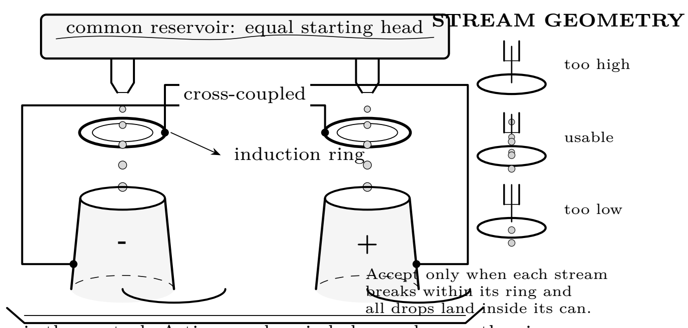
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Adult-built paired drip apparatus in a deep spill tray
- Two collector cans, two induction rings, and insulated cross-coupling
- Two adjustable plastic water lines with matched nozzles
- Low-capacitance electroscope with a fixed millimetre scale
- Stopwatch, ruler, two measuring cups, paper towels
- Designed high-resistance discharge lead, used only after flow stops

Predict first

Before water starts, choose one: will both electroscope leaves remain still, move together, or separate? What visible change would count as evidence for charge separation if sparks are forbidden? Predict which matters more: the total water delivered, or whether each stream breaks into droplets inside its induction ring.



Symmetry is the control. A tiny random imbalance chooses the signs; cross-coupling then reinforces it. No sign is assigned in advance.

Investigate

Adult preflight — water off

The learner observes from the marked boundary and does not touch the apparatus. The adult confirms all six items before proceeding:

1. Tray, frame, reservoir, cans, and rings are stable; the floor and nearby surfaces are dry. There are no exposed sharp edges.
2. The two cans are well separated, each ring is centred over its own can, and no loose wire can swing toward another conductor.
3. Each collector is connected only to the *opposite* ring. Trace both paths aloud. Same-side coupling is a wiring error.
4. The electroscope begins at zero and its scale can be read without anyone approaching the apparatus.
5. The intended high-resistance discharge lead is present, intact, and parked. It is not a hand-held improvised wire.
6. Stop immediately for a snap, visible spark, hair movement, tingling, unstable support, escaping water, or an electroscope reading above the chosen conservative stop mark.

Establish a hydraulic control

Ground and discharge the apparatus by its designed method. Run water for 10 seconds with charge unable to accumulate. Catch each stream separately, count its drops from slow-motion video or direct observation, and measure its volume. Adjust only the water controls until both streams break inside their rings, land centrally, and agree closely. Record the evidence; do not call "looks equal" a measurement.

Run	left drops/10 s	right drops/10 s	left mL/10 s	right mL/10 s
hydraulic control	________	________	________	________

Observe charge separation without discharge

Stop the water. Discharge through the intended high-resistance path, verify zero, then park the lead. Start the matched drips and the stopwatch together. From the boundary, read electroscope deflection against the fixed scale every five seconds. Stop at 20 seconds or at the preselected stop mark, whichever comes first. Repeat from a verified zero. A repeatable rise is evidence; one twitch is not.

Run	0 s	5 s	10 s	15 s	20 s
deflection, mm	______	______	______	______	______
repeat, mm	______	______	______	______	______

Return to a proved-safe state

Water off first. Wait for every drop to land. Apply the designed discharge path without touching either collector, hold it as specified for the apparatus, and verify the electroscope returns to zero. Only then may the adult approach. Dry the tray and floor before any adjustment.

Explain from the evidence

The hydraulic control shows that both sides began with similar water flow. During the isolated runs, a growing electroscope deflection shows that charge accumulated even though no battery was connected and no spark was allowed. A minute accidental imbalance polarizes the falling water. Each charged collector biases the opposite stream through its induction ring, so the imbalance reinforces itself. The falling water supplies the energy; the cross-coupling supplies positive feedback.

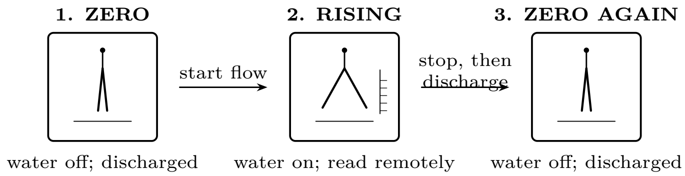
Interrogate the evidence. Did both repeated runs rise from zero? Was the direction consistent, or did the apparatus choose either sign? Could unequal drip rates explain the result? Which observation distinguishes feedback from a single charged object: continued growth, reversal after swapping a connection, or one initial jump?

Change one thing

Change only drip rate. After discharge and a verified zero, make both streams slightly slower while preserving their break points inside the rings. Repeat the same 20-second table. Compare the slope in millimetres per five seconds, not the maximum reading and never spark length.

If the trace stays flat, troubleshoot conservatively.

Observation	Water-off correction after discharge and zero verification
Both traces remain at zero	Trace opposite-side coupling; check that the electroscope responds to its approved test; dry insulators and tray.
One stream is continuous	Reduce flow until breakup occurs inside its ring; do not move a live ring.
Drops hit a ring or miss a can	Re-centre with water off; confirm supports cannot drift before restarting.
One run rises, the repeat does not	Restore the measured hydraulic control, dry all surfaces, and repeat from a verified zero.
Reading jumps or nears stop mark	Stop water immediately, discharge by the designed path, and increase separation before any further run.



The evidence is a complete state change, not merely a large reading: a verified neutral start, gradual deflection while drops fall, and a verified neutral finish after the prescribed shutdown.

Final proof. Accept the demonstration only if two isolated runs begin at zero, show a gradual measurable rise before the stop mark, make no spark or snap, keep every drop inside the collectors, and end at verified zero after high-resistance discharge. Otherwise record "not demonstrated" and stop.

Evidence record	
Prediction	______________________
One change	______________________
Observation	______________________
Claim + because	______________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.